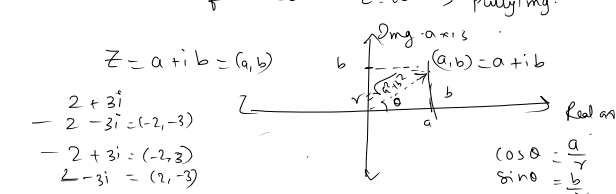


$z^2 + 1 = 0$
 $z^2 = -1$
 $z = \pm i$
 $i^2 = -1$
 $i^3 = i^2 \cdot i = -i$
 $i^4 = (i^2)^2 = 1$
 $i^{100} = (i^4)^{25} = 1^{25} = 1$
 $i^{123} = i^{120} \cdot i^3 = (i^4)^{30} \cdot (-i) = -i$

~~1~~
~~2~~
~~1-i~~

$z = a + ib$
 $a + ib = -2 + 3i$
 $a = -2, b = +3$

$z = a + ib$
 if $b = 0$ $z = a \rightarrow$ real no.
 if $a = 0$ $z = ib \rightarrow$ purely imag.



$z_1 = a_1 + ib_1$ $z_2 = a_2 + ib_2$

$z_1 \pm z_2 = (a_1 \pm a_2) + i(b_1 \pm b_2)$

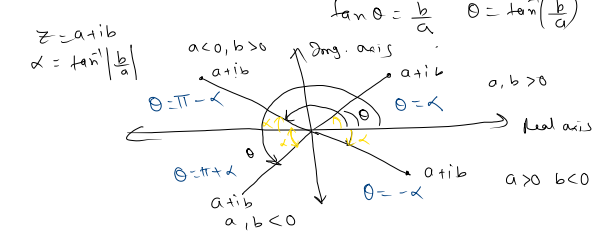
$(z_1 \cdot z_2) = (a_1 + ib_1)(a_2 + ib_2) = a_1 a_2 + ib_1 a_2 + ib_2 a_1 - b_1 b_2$
 $= (a_1 a_2 - b_1 b_2) + i(b_1 a_2 + b_2 a_1)$

$\frac{z_1}{z_2} = \frac{a_1 + ib_1}{a_2 + ib_2} \times \frac{a_2 - ib_2}{a_2 - ib_2}$

$z = a + ib$	$\bar{z}$
$2 + i$	$2 - i$
-2	-2
$-3 - i$	$-3 + i$
$-5i$	$5i$

Modulus: $z = a + ib$ $|z| = r = \sqrt{a^2 + b^2}$

Argument / Amplitude: $z = a + ib$
 $\cos \theta = \frac{a}{r}$ $\sin \theta = \frac{b}{r}$
 $\tan \theta = \frac{b}{a}$ $\theta = \tan^{-1}\left(\frac{b}{a}\right)$



$1 + i \Rightarrow \theta = \alpha = \pi/4$
 $1 - i \Rightarrow \theta = -\alpha = -\pi/4$
 $-1 + i \Rightarrow \theta = \pi - \alpha = \pi - \pi/4 = 3\pi/4$
 $-1 - i \Rightarrow \theta = \pi + \alpha = \pi + \pi/4 = 5\pi/4$

* forms of complex No.

$z = x + iy$
 $= r(\cos \theta + i \sin \theta)$

$x = r \cos \theta$ $y = r \sin \theta$
 Polar form

$= r e^{i\theta}$

$e^{i\theta} = \cos \theta + i \sin \theta$

$z = 1 + i = \sqrt{2} e^{i\pi/4}$
 $\bar{z} = 1 - i = \sqrt{2} e^{-i\pi/4}$
 $-1 + i = \sqrt{2} e^{i3\pi/4}$
 $-1 - i = \sqrt{2} e^{i5\pi/4}$

$r = |z| = \sqrt{x^2 + y^2}$
 $|z| = |\bar{z}| = |-z|$

Conjugate of $z = \overline{z}$

$$z = a + ib$$

$$\overline{z} = a - ib$$

Argand's Diagram

$$z = x + iy$$

$$\text{modulus } z = |z| = \sqrt{x^2 + y^2} = r$$

$$x = r \cos \theta$$

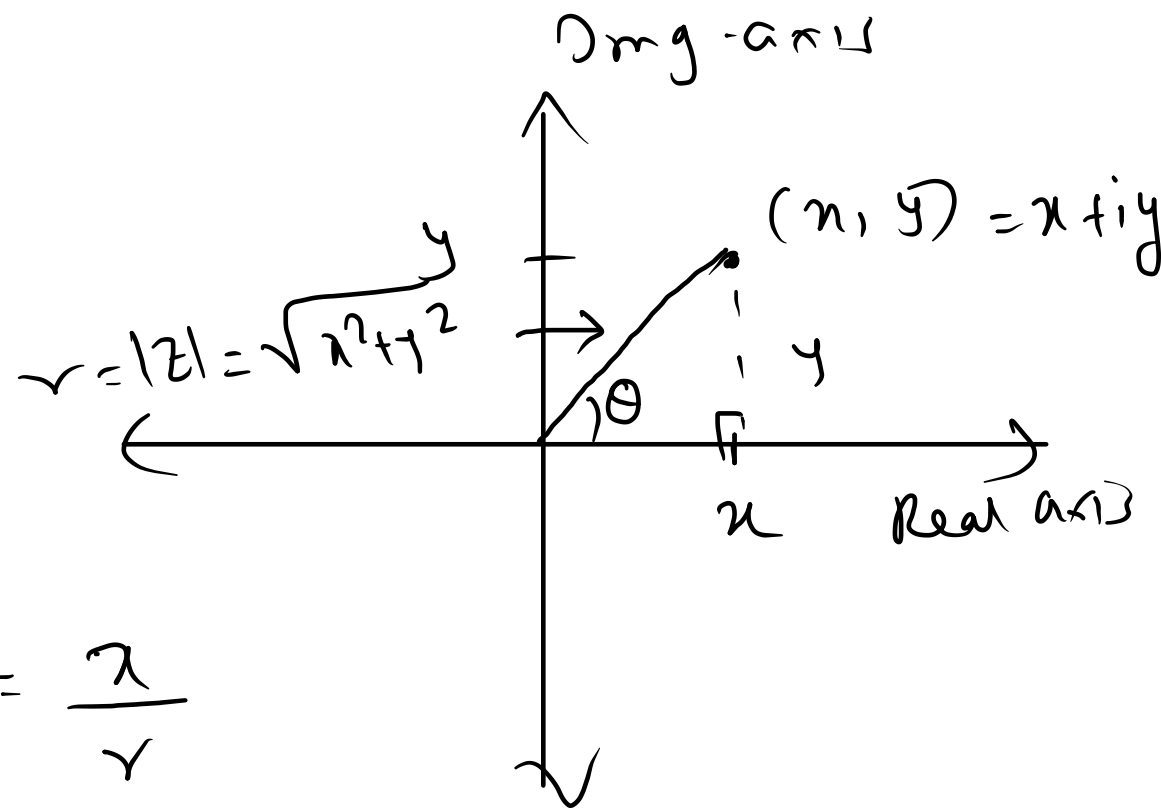
$$y = r \sin \theta$$

$$\tan \theta = \frac{y}{x}$$

$$\cos \theta = \frac{x}{r}$$

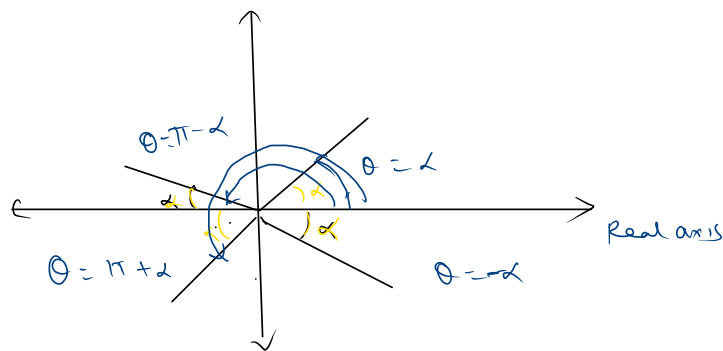
$$\sin \theta = \frac{y}{r}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \text{argument} - \text{the value bet}^n -\pi \& \pi$$



* principal value
of θ is

$$\alpha = \tan^{-1} \left| \frac{y}{x} \right|$$

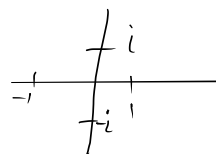


z		θ
$x+iy$	$x, y > 0$	$\theta = \alpha$
$x+iy$	$x < 0, y > 0$	$\theta = \pi - \alpha$
$x+iy$	$x, y < 0$	$\theta = \pi + \alpha$
$x+iy$	$x > 0, y < 0$	$\theta = -\alpha$

$$|z| = |-z| = |\bar{z}|$$

$\alpha = \tan^{-1} \left \frac{y}{x} \right $			
$= \frac{\pi}{4}$	$1+i$	$r = z $	$\alpha = \frac{\pi}{4}$
	$1-i$	$\sqrt{2}$	$\theta = \alpha = \frac{\pi}{4}$
	$-1+i$	$\sqrt{2}$	$\theta = -\alpha = -\frac{\pi}{4}$
	$-1-i$	$\sqrt{2}$	$\theta = \pi - \alpha = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$
			$\theta = \pi + \alpha = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

	r	θ
1	1	0
-1	1	π
i	1	$\frac{\pi}{2}$
$-i$	1	$-\frac{\pi}{2}$



$$Z = x+iy \rightarrow \text{Cartesian form}$$

$$= r(\cos \theta + i \sin \theta) \quad \begin{matrix} x = r \cos \theta & y = r \sin \theta \end{matrix}$$

Polar form

$$= r e^{i\theta} \rightarrow \text{exponential form}$$

Euler's Identity

$$e^{i\theta} = \cos \theta + i \sin \theta \quad e^{i\theta} + e^{-i\theta} = 2 \cos \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta \quad e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

$$Z = r e^{i\theta} \quad \frac{1}{Z} = \frac{1}{r} e^{-i\theta}$$

$$\bar{Z} = r e^{-i\theta}$$

De Moivre's Thm
for any Rational no. n .

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$(\cos \theta + i \sin \theta)^{-n} = \cos n\theta - i \sin n\theta.$$

$$= (\cos \theta - i \sin \theta)^n$$

$$(e^{i\theta})^n = \underline{\underline{e^{in\theta}}}$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

Note: $(\cos \phi + i \sin \theta)^n \neq \cos n\phi + i \sin n\theta$

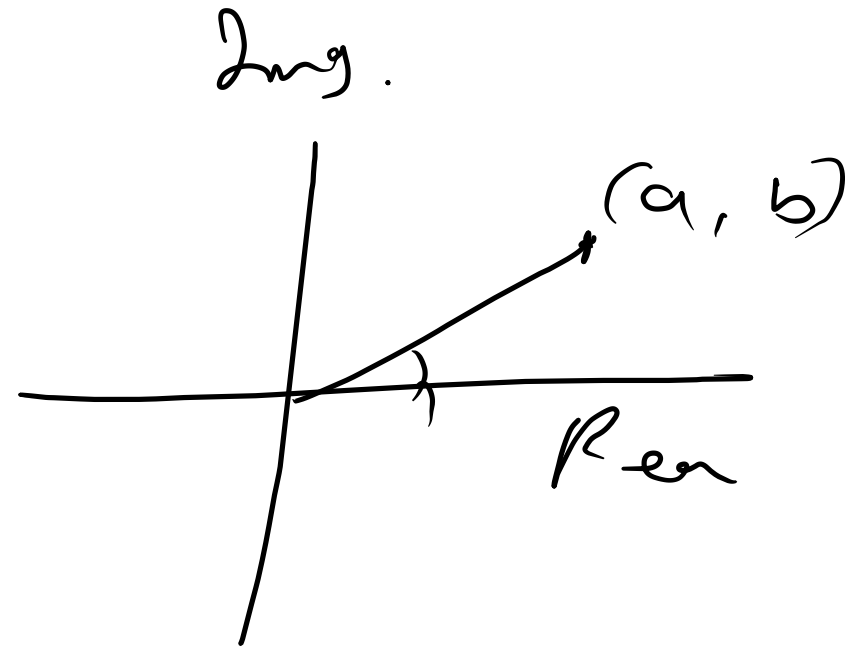
$$(\sin \theta + i \cos \theta)^n \neq \sin n\theta + i \cos n\theta$$

$$\begin{aligned} (\sin \theta + i \cos \theta)^n &= \left[\cos\left(\frac{\pi}{2} - \theta\right) + i \sin\left(\frac{\pi}{2} - \theta\right) \right]^n \\ &= \cos n\left(\frac{\pi}{2} - \theta\right) + i \sin n\left(\frac{\pi}{2} - \theta\right) \end{aligned}$$

$$Z = \textcircled{a} + i \textcircled{b}$$

$$|Z| = \sqrt{a^2 + b^2}$$

$$Z = x + iy$$



modulus

$$|z| = |\bar{z}| = |-z| \quad \checkmark$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad z_2 \neq 0$$

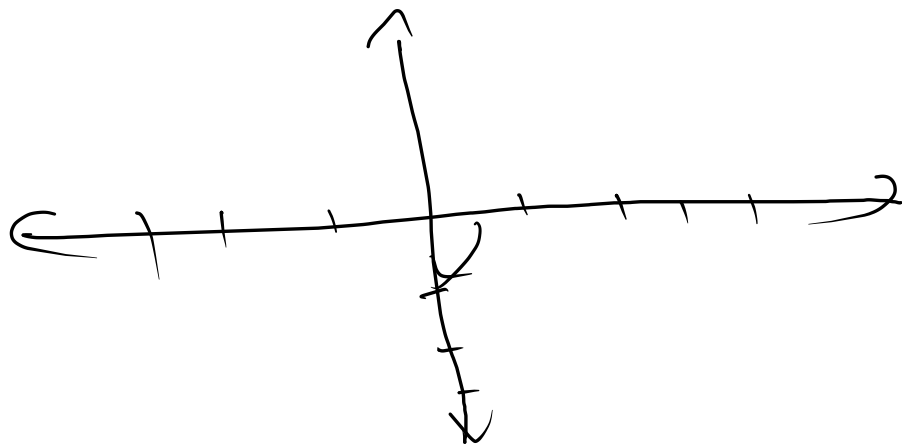
argument

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2}$$

$$r_1 r_2 e^{i(\theta_1 + \theta_2)}$$



$$z = x + iy$$

$$\bar{z} = x - iy$$

$$z + \bar{z} = 2x \Rightarrow$$

$$x = \operatorname{Re}(z) = \frac{z + \bar{z}}{2}$$

$$z - \bar{z} = 2iy \Rightarrow y = \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

z	θ	$re^{i\theta}$
1	0	e^{i0}
i	$\pi/2$	$e^{i\pi/2}$
-1	π	$e^{i\pi}$
$-i$	$-\pi/2$	$e^{-i\pi/2}$

$$z = 10i \quad |z| = 10$$

$$z = -10i \quad |z| = 10$$

$$z = -5i \quad |z| = 5$$

Prove that

$$\frac{(1+i)^8(\sqrt{3}-i)^4}{(1-i)^4(\sqrt{3}+i)^8} = -\frac{1}{4}$$

$$1+i \quad r = \sqrt{2}$$

$$\theta = \frac{\pi}{4}$$

$$\sqrt{3}+i \quad 2$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$1+i = \sqrt{2} e^{i\pi/4}$$

$$1-i = \sqrt{2} e^{-i\pi/4}$$

$$\sqrt{3}+i = 2 e^{i\pi/6}$$

$$\sqrt{3}-i = 2 e^{-i\pi/6}$$

$$\frac{(1+i)^8 (\sqrt{3}-i)^4}{(1-i)^4 (\sqrt{3}+i)^8} = \frac{(\sqrt{2})^8 (e^{i\pi/4})^8 \cancel{(2)^4} (e^{-i\pi/6})^4}{\cancel{(\sqrt{2})^4} (e^{-i\pi/4})^4 (2)^8 (e^{i\pi/6})^8}$$

$$= \frac{(\sqrt{2})^4 e^{i2\pi}}{e^{-i\pi}} \frac{e^{-i2\pi/3}}{2^4 e^{i4\pi/3}}$$

$$= \frac{4}{16} e^{i3\pi} e^{-i2\pi} = \frac{1}{4} e^{i\pi}$$

$$= \frac{1}{4} (\cos \pi + i \sin \pi)$$

$$= \frac{1}{4} (-1)$$

For n integer

$$\sin n\pi = 0$$

$$\cos n\pi = (-1)^n$$

De Moivre's Thm.

For any rational no. n .

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$(e^{i\theta})^n = e^{in\theta}$$

Note: ① $(\cos \theta_1 + i \sin \theta_2)^n \neq \cos n\theta_1 + i \sin n\theta_2$

② $(\sin \theta + i \cos \theta)^n \neq \sin n\theta + i \cos n\theta$

$$\begin{aligned} (\sin \theta + i \cos \theta)^n &= \left[\cos\left(\frac{\pi}{2} - \theta\right) + i \sin\left(\frac{\pi}{2} - \theta\right) \right]^n \\ &= \cos n\left(\frac{\pi}{2} - \theta\right) + i \sin n\left(\frac{\pi}{2} - \theta\right) \end{aligned}$$

③ $z = \cos \theta + i \sin \theta$

$$\begin{aligned} \frac{1}{z} = z^{-1} &= (\cos \theta + i \sin \theta)^{-1} = \cos(-\theta) + i \sin(-\theta) \\ &= \cos \theta - i \sin \theta \end{aligned}$$

$$\begin{aligned} (\cos \theta + i \sin \theta)^{-n} &= \cos n\theta - i \sin n\theta \\ &= (\cos \theta - i \sin \theta)^n \end{aligned}$$

- Find the modulus and the principal value of the

- argument of $\frac{(1+i\sqrt{3})^{16}}{(\sqrt{3}-i)^{17}}$

$$1+i\sqrt{3} = 2 e^{i\pi/3}$$

$$\sqrt{3}-i = 2 e^{-i\pi/6}$$

$$\cos(2k\pi + \theta) = \cos \theta$$

$$\sin(2k\pi + \theta) = \sin \theta$$

$$\frac{(1+i\sqrt{3})^{16}}{(\sqrt{3}-i)^{17}} = \frac{(2 e^{i\pi/3})^{16}}{(2 e^{-i\pi/6})^{17}}$$

$$= \frac{1}{2} \frac{e^{i16\pi/3}}{e^{-i17\pi/6}}$$

$$= \frac{1}{2} e^{i\left(\frac{16\pi}{3} + \frac{17\pi}{6}\right)}$$

$$= \frac{1}{2} e^{i\left(\frac{49\pi}{6}\right)}$$

modulus of $\frac{(1+i\sqrt{3})^{16}}{(\sqrt{3}-i)^{17}}$ is $\frac{1}{2}$ & argument is $\left(\frac{49\pi}{6}\right)$

$$\frac{(1+i\sqrt{3})^{16}}{(\sqrt{3}-i)^{17}} = \frac{1}{2} \left[\cos\left(\frac{49\pi}{6}\right) + i \sin\left(\frac{49\pi}{6}\right) \right]$$

$$= \frac{1}{2} \left[\cos\left(8\pi + \frac{\pi}{6}\right) + i \sin\left(8\pi + \frac{\pi}{6}\right) \right]$$

$$= \frac{1}{2} \left[\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right] \quad \left| \begin{array}{l} \text{principal value} \\ \text{of arg. is } \frac{\pi}{6} \end{array} \right.$$

- If $z = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$ and $\bar{z}$ is the conjugate of z prove that $(z)^{10} + (\bar{z})^{10} = 0$.

$$z = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$$

$$= e^{i\pi/4}$$

$$\bar{z} = e^{-i\pi/4}$$

$$r = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$$

$$\theta = \alpha = \pi/4$$

$$z^{10} + \bar{z}^{10} = \left(e^{i\pi/4}\right)^{10} + \left(e^{-i\pi/4}\right)^{10} = e^{i5\pi/2} + e^{-i5\pi/2}$$

$$= \cos\left(\frac{5\pi}{2}\right) + i\sin\left(\frac{5\pi}{2}\right) + \cos\left(\frac{5\pi}{2}\right) - i\sin\left(\frac{5\pi}{2}\right)$$

$$= 2\cos\left(\frac{5\pi}{2}\right) = 2\cos\left(2\pi + \frac{\pi}{2}\right) = 2\cos\frac{\pi}{2} = 0$$

• If α, β are the roots of the equation $x^2 - 2x + 2 = 0$,
prove that

• $\alpha^n + \beta^n = 2 \cdot 2^{n/2} \cos n\pi/4$,

• Hence, deduce that $\alpha^8 + \beta^8 = 32$

$$x = 1 \pm i$$

$$\alpha = 1 + i \quad \beta = 1 - i$$

$$= \sqrt{2} e^{i\pi/4} \quad = \sqrt{2} e^{-i\pi/4}$$

$$\begin{aligned} \alpha^n + \beta^n &= (\sqrt{2} e^{i\pi/4})^n + (\sqrt{2} e^{-i\pi/4})^n \\ &= (\sqrt{2})^n \left[e^{i n \pi / 4} + e^{-i n \pi / 4} \right] \\ &= 2^{n/2} \left(\cos\left(\frac{n\pi}{4}\right) + i \sin\left(\frac{n\pi}{4}\right) + \cos\left(\frac{n\pi}{4}\right) - i \sin\left(\frac{n\pi}{4}\right) \right) \\ &= 2^{n/2} \cdot 2 \cos\left(\frac{n\pi}{4}\right) \end{aligned}$$

put $n=8$

$$\alpha^8 + \beta^8 = 2^4 \cdot 2 \cos\left(\frac{8\pi}{4}\right) = 2^5 \cos 2\pi = 32 //$$

Hw

α, β are roots of $x^2 - 2\sqrt{3}x + 4 = 0$
p.t. $\alpha^3 + \beta^3 = 0$ & $\alpha^3 - \beta^3 = 16i$

Example ..

- If $x - \frac{1}{x} = 2i \sin \theta$, $y - \frac{1}{y} = 2i \sin \phi$, $z - \frac{1}{z} = 2i \sin \psi$, prove that
- ① $xyz + \frac{1}{xyz} = 2 \cos(\theta + \phi + \psi)$ ② $\frac{\sqrt[m]{x}}{\sqrt[n]{y}} + \frac{\sqrt[n]{y}}{\sqrt[m]{x}} = 2 \cos\left(\frac{\theta}{m} - \frac{\phi}{n}\right)$

$$x - \frac{1}{x} = 2i \sin \theta$$

$$x^2 - 1 = (2i \sin \theta)x \Rightarrow x^2 - (2i \sin \theta)x - 1 = 0$$

$$a = 1, b = -2i \sin \theta, c = -1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2i \sin \theta \pm \sqrt{4 \sin^2 \theta + 4}}{2}$$

$$= \frac{2i \sin \theta \pm 2 \sqrt{1 - \sin^2 \theta}}{2}$$

$$= i \sin \theta \pm \cos \theta$$

$$= \pm \cos \theta + i \sin \theta$$

Consider $x = \cos \theta + i \sin \theta = e^{i\theta}$

Similarly $y = \cos \phi + i \sin \phi = e^{i\phi}$

$$z = \cos \psi + i \sin \psi = e^{i\psi}$$

$$xyz + \frac{1}{xyz} = e^{i\theta} \cdot e^{i\phi} \cdot e^{i\psi} + \frac{1}{e^{i\theta} \cdot e^{i\phi} \cdot e^{i\psi}}$$

$$= e^{i(\theta + \phi + \psi)} + e^{-i(\theta + \phi + \psi)}$$

$$= \cos(\theta + \phi + \psi) + i \sin(\theta + \phi + \psi)$$

$$+ \cos(\theta + \phi + \psi) - i \sin(\theta + \phi + \psi)$$

$$= 2 \cos(\theta + \phi + \psi)$$

$$\sqrt[m]{x}$$

$$= (x)^{1/m}$$

- If $x_r = \cos \frac{\pi}{3^r} + i \sin \frac{\pi}{3^r}$, prove that (i) $x_1 x_2 x_3 \dots \text{ad. inf.} = i$ (ii) $x_0 x_1 x_2 \dots \text{ad. inf.} = -i$

$$x_r = \cos \frac{\pi}{3^r} + i \sin \frac{\pi}{3^r} = e^{i \pi / 3^r}$$

$$\left| 1 + r + r^2 + \dots \right| = \frac{1}{1-r} \quad |r| < 1$$

$$x_1 \cdot x_2 \cdot x_3 \dots \infty$$

$$= e^{i \pi / 3} \cdot e^{i \pi / 3^2} \cdot e^{i \pi / 3^3} \dots \infty$$

$$= e^{(i \pi / 3 + i \pi / 3^2 + i \pi / 3^3 + \dots)} = e^{i \pi / 3 (1 + \frac{1}{3} + \frac{1}{3^2} + \dots)}$$

$$= e^{i \pi / 3 \left(\frac{1}{1 - 1/3} \right)} = e^{i \pi / 3 (3/2)}$$

$$= e^{i \pi / 2}$$

$$= \cos \pi / 2 + i \sin \pi / 2 = 0 + i = i$$

$$x_0 (x_1 x_2 \dots \infty)$$

$$= e^{i \pi} (i) = (\cos \pi + i \sin \pi) i$$

$$= (-1) i = -i$$

